Graphical Abstract

Automated Compliance Checking in AEC: A Conceptual Framework with Systematic Review and Automated Thematic Synthesis (2022–2025)

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Highlights

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- Conceptual Framework with Systematic Review and Automated Thematic Synthesis of ACC in the AEC from 2022 to 2025.
- Development and implementation of an NLP-enhanced semantic embedding (MiniLM-cosine similarity) research framework for identifying and analysing thematic trends.

Automated Compliance Checking in AEC: A Conceptual Framework with Systematic Review and Automated Thematic Synthesis (2022–2025)

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Abstract

Automated compliance checking (ACC) is transforming the way architects, engineers, and constructors demonstrate code and standards conformance. This is a 2022–2025 systematic review and thematic synthesis of ACC in Architecture, Engineering, and Construction (AEC), uniformly sampled from Scopus, Web of Science, IEEE Xplore, and a deep search using ChatGPT. We proposed a conceptual framework in the AEC field and, fitting the corpus into ten themes that include Multimodal Information Extraction, Formalisation, BIM/IFC Alignment, Ontologies and Knowledge Graphs, Rule Representation, Explainability, Human-in-the-Loop, Model–Driven Compliance Intelligence, Benchmarking, and Tool Deployment, we first created a Python de-duplication routine to clean the data. After that, we conducted theme coverage analysis following relevance screening, and performed semantic ranking using MiniLM embeddings. After that, we created a matrix covering the highest-ranked papers. For each, we read and summarised the authors, goals, tools, domain, theme mapping, and a binary relevance flag (Yes/No). Of the 33 papers, three were judged not relevant to the assigned theme and were re-mapped accordingly.

Overall, the review shows strong progress in upstream tasks; notably formalisation and rule representation, but weaker integration into production workflows and limited attention to explainability and human-in-the-loop oversight. Most systems are built from three recurring execution paradigms: ontology over SWRL inference, SHACL or SPARQL validation, and procedural checks. Evaluation commonly uses expert gold standards with precision, recall, F1 and time-to-result as headline metrics. The key gaps are end-to-end interoperability across tools and platforms, standardised, human-readable failure explanations, and demonstrated breadth of coverage across codes and project types. Addressing these will require embedding semantic methods-including Large Language Models and classic NLP into standards-based tools, establishing shared open benchmarks and testbeds, and running large-scale, multidisciplinary pilots.

Keywords: Automated Compliance Checking (ACC), Architecture Engineering and Construction (AEC), Thematic Literature Review, Research Framework, MiniLM Embeddings, Cosine Similarity.

1. Introduction

Maintaining building projects to the statutory requirements is core to safety, sustainability, and quality in general within the Architecture, Engineering and Construction (AEC) industry. This conformity has traditionally been check-marked for manual reviews by professionals or experts. Such reviews, although essential over extended periods, are prone to human error, require substantial time and resources, and often result in program delays and increased costs [1, 2]. The increasing sophistication of modern regulations and a focus on designing in a sustainable manner has revealed the shortcomings of strictly manual processes [3].

Automated Compliance Checking (ACC) has thus emerged as a feasible alternative. The ACC platforms perform an automatic comparison of the digital building designs to the applicable codes and standards, which is supported by the rich and object-oriented data environment provided by Building Information Modelling (BIM) [4, 5]. Interoperability and trustworthy data exchange, which are the preconditions to successful automation, are also promoted by the adoption of the Industry Foundation Classes (IFC) open standard that is widespread [6].

The development of ACC has been in defined waves. The initial systems, such as the CORENET ePlanCheck in Singapore and the DesignCheck in Australia, integrated all the regulatory provisions into procedure code. This approach offered rapid run-times and was closely to printed regulations. Nevertheless, any slight changes in legislations require a programmer to rewrite and recompile the underlying collection of libraries, which would soon be labour intensive and difficult to maintain [4]. It was confirmed by researchers of their long-term performance that the hard-coded strategy would not be able to be extended beyond a small rule set without significant redevelopment [7].

Researchers have adopted the use of ontology-based frameworks to overcome these limitations. In this case, regulatory concepts (e.g., storey, exit width, travel distance) are formalised as classes and properties using the Web Ontology Language (OWL). Constraints are typically expressed declaratively in the Semantic Web Rule Language (SWRL) or in the Protocol and RDF Query Language (SPARQL), enabling a reasoning engine such as Apache Jena to infer which elements in a BIM model are compliant. By decoupling knowledge from executable code, this architecture makes it possible for domain experts to revise rules directly, without editing source files. At the same time, the ontology can be reused across different projects and application domains [1, 8].

The most recent wave of ACC studies turned to Natural Language Processing (NLP) and Machine Learning (ML) to solve the problem of rule capture. A common workflow begins by tokenising and parsing the original legal text. Named-entity recognition by domain-specific named-entities is then used to detect key actors (e.g., door, stair, refuge) and quantitative parameters. Such elements are projected on a regulatory ontology and each obligation can be represented as a formal triple and then be encoded as a SWRL axiom or a statement in the Shapes Constraint Language (SHACL) [9, 10]. Complementary Machine Learning (ML) classifiers are often used to detect clause type (requirement, exception, applicability) and to normalise units, thereby reducing manual post-editing.

Recent hybrid researches integrate NLP-based rule-extraction pipelines, semantic-web repositories expressed in OWL over Resource Description Framework (RDF), and data-driven models trained on precedents drawn from multiple jurisdictions [11, 9]. This approaches produce more transferable rule bases and much less fragile, in the case of changes to the regulations, or changes in wording or translation [11]. Building on that foundation, Artificial Intelligence (AI) and Large Language Models (LLM) finetuned on regulatory corpora have shown to be able to generate candidate rules, propose plausible preconditions, and complete gaps in BIM ontologies [12]. Experimental results in early works show promising rates of recall and coverage, yet the techniques remain constrained by opaque reasoning paths and the paucity of high-quality, openly accessible regulatory datasets [13, 14]. These shortcomings will have to be overcome before LLM-generated rules can be relied upon in safety-critical compliance workflows [13, 15].

Although the technical state of the art has advanced markedly, the large-scale deployment of automated compliance checking (ACC) in everyday practice is still limited. Four inter-related obstacles dominate the discourse:

- **Rule-capture bottleneck:** Converting lengthy, sometimes ambiguous legal prose into machine-interpretable constraints continues to rely heavily on manual expertise, even though NLP pipelines have begun to extract entities, thresholds, and logical conditions with increasing accuracy [9, 10, 13].

- **Limited semantic interoperability:** Mainstream BIM models typically contain geometry with a small set of attributes; they rarely include the functional or contextual semantics needed for thorough, multi-disciplinary checks, and systematic linkage to external knowledge bases is still the exception rather than the rule [11, 6].

- **Domain-specific operational solutions:** Commercial and research prototypes alike tend to focus on one regulatory theme such as fire safety, energy, accessibility, or similar and their rule bases cannot readily be ported to other domains without extensive re-engineering [4, 3].

- **Opaque inference engines:** The inference engines embedded in popular checkers (e.g., Solibri or various Revit plug-ins) are opaque; their proprietary code precludes independent audit of the reasoning path and, in consequence, limits stakeholder trust [7, 3].

Past researches and surveys have considered individual aspects of automated compliance checking, e.g. BIM-centred checking processes [5], ontological reasoning [1], or language-rule interpretation pipelines [9, 10]. These surveys are mostly narrative and siloed, as such they only reveal aspects of a growing complex research environment. Even the latest synthesis, like that of Yamusa et al. [2], although effectively expanding the lens to knowledge-graph reasoning and early experiments involving LLMs, nevertheless does not cover the panorama of the post-2022 growth of AI, NLP, and semantics-focused ACC studies in a data-driven and cross-theme way.

This gap is filled by the current review, which provides a theme-based synthesis of the published work during the period of 2022-2025 in a computationally supported form. Based on MiniLM sentence embeddings, cosine-similarity, and visual analytics, we can map the corpus and to uncover latent interconnections as well as to detect emerging research fronts. This evidence-based practice presents the first quantitative account of the reconfiguring of the ACC field through AI, NLP, and hybrid semantic-learning methods.

Therefore, four research questions guide the review:

1. How are today's studies turning narrative regulations into machine-readable rules, linking them to BIM/IFC data, and which representation/reasoning patterns dominate?
2. To what extent do multimodal extraction pipelines, ontologies, knowledge graphs and other hybrid AI techniques underpin current ACC research?
3. How are Model–Driven Compliance Intelligence and human-in-the-loop strategies reshaping the end-to-end compliance-checking workflow?
4. What methods are emerging to guarantee interoperability, explainability, benchmarking and stakeholder confidence, and how are these being embedded in practical ACC tools?

This paper is consists of five sections. In section 2, the research methodology is explained, the steps that were followed in collecting the data, searching of databases, and the screening and analytical procedures that were followed systematically are explained. In section 3, a thorough thematic synthesis is conducted, where previous literature is synthesised into ten thematic strands that are interrelated. Section 4 presents the discussion and implications of the identified themes, analyses the areas where conceptual or methodological gaps remain, and frames these findings in the broader context of the discussion of the automated compliance checking. The conclusion, which concludes the study by summarising its most important contributions and listing the priority directions of the future research and practical application, is provided in Section 5.

2. Research Methodology

2.1. Literature Search Process

We chose to follow a PRISMA search strategy with four complementary sources of information to maximise both disciplinary breadths before starting the review [16]. Scopus, Web Of Science, IEEE Xplore were consulted to retrieve bibliographic records that index most of the high-quality, peer-reviewed literature in the AEC and computing fields. To further reduce

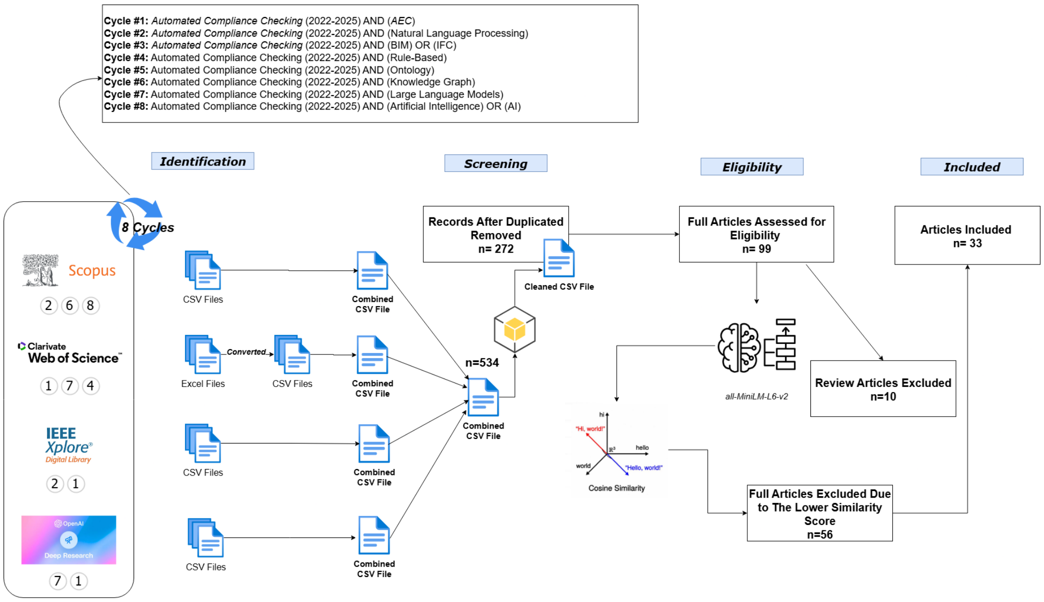

Figure 1: PRISMA Flow Diagram

publication bias and to include the most recent and indexed work, we added these databases with Deep Search through an LLM assisted Deep Search via ChatGPT. This search feature searches in the three databases together to verify the search domain beyond the mere indexing services and guarantees that the most current contributions were captured in our corpus.

2.2. Search Strategy and Query Design

Since the construct of ACC is a cross-disciplinary, we used an eight-cycle query scheme that was iterative as illustrated in Figure 1. Each cycle constrained the core phrase "Automated Compliance Checking" to the most recent $\approx$ four-year window (2022-2025) and appended one thematic lens to capture domain-specific vocabularies or emerging technologies as shown in Table 1.

Table 1: Keywords and Justification for ACC Literature Review.

Cycle	Boolean expression (Title, Abstract, Keywords)	Justification
#1	ACC AND *"AEC"*	General construction context
#2	ACC AND *"Natural Language Processing"*	Linguistic rule extraction trend
#3	ACC AND (*"BIM"* OR *"IFC"*)	Model-centric ACC research
#4	ACC AND *"Rule-Based"*	Traditional expert-system approaches
#5	ACC AND *"Ontology"*	Semantic-web approaches
#6	ACC AND *"Knowledge Graph"*	Graph-based reasoning methods
#7	ACC AND *"Large Language Model"*	Recent Gen-AI applications
#8	ACC AND (*"Artificial Intelligence"* OR *"AI"*)	Broader AI innovations

2.3. Screening and Eligibility Criteria

The bibliographic exports retrieved at the four sources of information differed significantly in terms of the type of file they accessed and the way their metadata were put into place. A Python workflow was created to automate the process and run in two stages based on a two-fold Python workflow to unify these diverse outputs.

2.3.1. File Standardisation and Intra-Source Deduplication:

Web of Science (WoS) export their output in excel sheets (`.xls`/`.xlsx`) but Scopus, IEEE Xplore and ChatGPT Deep Search were exporting outputs in comma-separated values (CSV) files with their own field format. We wanted to counter these outputs by having our own Python program which would perform five repetitive actions. Any file, no matter what relayed, were converted on the basis of UTF-8-encoded CSV formats so as to make them congenial to usage on any platform. Secondly, the most popular encoding artefacts were voluntarily corrected (e.g. "UKâ€™s"). Third, all databases that were used by the script exported many times and combined all of them in one intermediary file that stored all the queries that were run in any of the sources. Fourth, Pandas dropped intra-source duplicates using `drop_duplicates()` method which deletes the duplicates that repeat one to the other per column. Lastly, the output of individual sources that have been cleaned was saved directly on disk in the form of standardised format of names (e.g., `Scopus_combined_unique.csv`). The result of this step was the validated, deduplicated data set by database, and the equivalent platform to proceed to the next step that is the cross-source integration.

2.3.2. Schema Harmonisation and Cross-Source Integration:

In the second step, we matched the four files that are source specific and combined them into the single corpus. The unique column names of the databases; *Document Title* in IEEE Xplore, *Article Title* in Web of Science, etc were converted into a standardized five column schema `Author`, `Title`, `Year`, `Keywords`, and `Abstract`. Once the data had been normalised, the data were concatenated and finally a Pandas `drop_duplicates()` was then applied to column `Title` eradicating duplicates across sources. And finally, all the remaining white-space has been removed in the text fields and the whole integrated data set is written out to disk in a single file `AllSources_Combined_Clean_Final.csv` as a single, de-duplicated source of data any further kind of processing operations can be done subsequently.

2.3.3. Manual Relevance Screening:

The process of automated integration involved the generation of 272 unique bibliographic records. The two-step relevance appraisal was then carried out by the authors. Initial screening was performed first by title and abstract to be screened against the predefined inclusion criteria and exclude commentaries, editorials, and studies outside automated compliance checking in the AEC field. Second, the complete texts of the rest of the records were reviewed to verify the methodological rigour and substantive orientations to definite contributions in the fields of empirical or technical descriptions. This consecutive sifting narrowed down the records into 99 articles, the base of the evidence of the current review.

2.4. Conceptual Framework of ACC in ACE

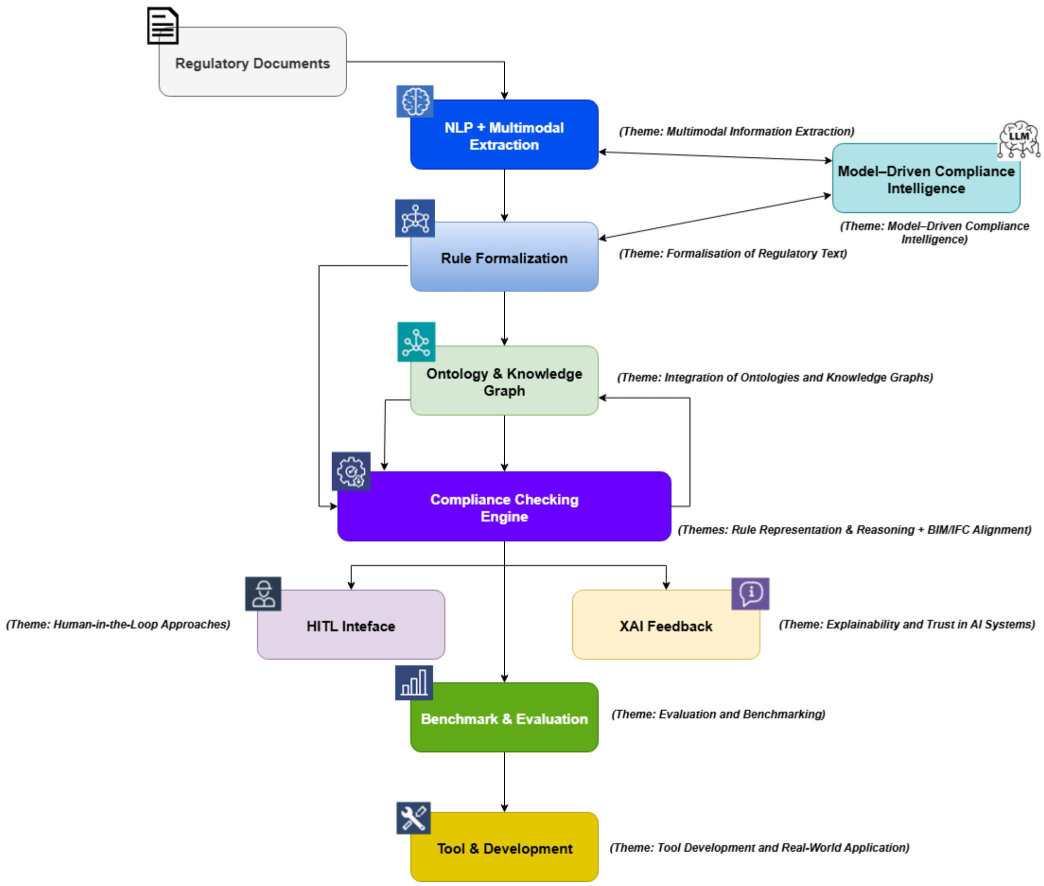

Figure 2: Conceptual Framework of ACC in ACE

Figure 2 provides the conceptual framework of an ACC in AEC domain. This figures the functional parts of a compliance system working end-to-end. It is on this pipeline that we ascribe the ten thematic research that we have employed in our semantic analysis.

The development will start with the Regulatory Documents being the key source of compliance requirements. The documents themselves are heterogeneous in nature; mixing natural language with tabular schedules and visual diagrams and so requires both NLP and Multimodal Extraction to convert the unstructured text into structured data. The research efforts [9, 19] have proved the usefulness of multimodal extraction methods in extracting the regulatory clauses and the constraints on the parameters given text and graphics.

After relevant clauses are pulled out, they must be patterned into logic that is comprehensible to computer-a stage referred to as Rule Formalisation. In this case, the textual provisions are transformed into semantic rules or constraints to have them being handled by automated reasoning engines. Preliminary studies demonstrated that semantic and template-based approaches are practical, although most of the rules are designed to be understood by humans, in particular, a semantic approach would not be able to extract regulations listed as legal categories, as they exist in a written form [20, 21].

To support interoperability and enhance semantics, a vast number of researchers store the set of formalised rules interior to the Ontology or Knowledge Graph (KG). The regulatory concepts are translated into BIM objects and IFC classes with shared conceptual models, establishing a standardized language of compliance tools [11, 22]. The backbones are used to support checking rules engines that are used to check BIM models or foster digital twins against rule codifications [23]. The precision of the checks depends on the closeness of the correlation between the BIM/IFC which means making sure that the domain-specific attributes and geometric restrictions are checked correspondingly.

Together with full automation, Human-in-the-Loop (HITL) interfaces enable expert monitoring on ambiguous situations and refinement of the rules in small increments [24, 25]. Likewise, Explainable AI (XAI) feedback systems serve to answer the increasing need to know and trust AI-based compliance systems that facilitate auditing and accountability [26, 27].

Lastly, Benchmarking and Evaluation makes tools and frameworks robust in methodological terms via reproducible measures [27, 28], and Tool Deployment brings applicability to the research prototypes by rendering them operational and adapting workflows in the AEC industry [28].

The conceptual framework pipeline is complemented by new trends that provide an opportunity to extend the arguments through reasoning especially with applications of unstructured legal data and an opportunity to customize arguments based on queries to the data and adapt to the setting **LLMs** [29].

The ten thematic areas delineated in Table 2, the pipeline may be considered as a conceptual framework which might have been applied to the literature study, as well as a road map or a guide on how the future study of enhanced ACC systems will take.

Theme	Description
Multimodal Information Extraction	Extracting regulatory data from text, images, tables, diagrams using NLP and computer vision.
Formalisation of Regulatory Text	Transforming legal/regulatory text into structured, machine-readable formats via parsing, modelling, and logic encoding.
Semantic Alignment with BIM/IFC	Mapping regulatory requirements to BIM objects using IFC and buildingSMART standards.
Integration of Ontologies and Knowledge Graphs	Using ontologies and knowledge graphs to model regulatory concepts, relations, and rule hierarchies.
Rule Representation and Reasoning	Expressing compliance rules with formal languages and enabling inference.
Model–Driven Compliance Intelligence	Applying LLMs and Retrieval-Augmented Generation for compliance understanding and knowledge augmentation.
Explainability and Trust in AI Systems	Ensuring AI compliance tools offer transparent, interpretable explanations for trust and auditability.
Human-in-the-Loop Approaches	Incorporating human experts into compliance pipelines for supervision and refinement.
Evaluation and Benchmarking	Validating ACC systems with benchmark datasets, metrics, and comparative performance testing.
Tool Development and Real-World Application	Developing and deploying ACC tools within real-world AEC workflows and systems.

Table 2: Themes and their descriptions for ACC research

2.5. Semantic Filtering and Scoring

After finalising, creating and downloading all the PDFs in the articles dataset, we converted each article by formatting them to one machine readable text file which contained the text, and text of the tables in each article and the OCR text which was in figures and diagrams. That is, we could embed those representatives of rich text, but in fact we also can embed each sentence with the *all-MiniLM-L6-v2* model from the Sentence-Transformers framework, which is based on the architecture introduced by Reimers and Gurevych [17], and that gives us now convenient 384-dimensional vectors with which to measure the meaning of that sentence. Then we computed the cosine similarity as a metric score of the similarity between those vectors and a bank of theme specific, so-called reference embeddings, for a well-known metric of measuring cosine similarity in a high dimensional space [18]. With such scores of similarities we can rank the papers on the basis of how similar each of them followed each theme, and after getting the scores of similarities, we will have an automatically ranked reading list. Since, through contextual encoding, the approach at detecting conceptually similar work that is inadmissible into keyword-based filtering, the approach is synced with or predating the review practices that employ NLP in AEC [9]. Figure 3 illustrates the comprehensive workflow of the research methodology.

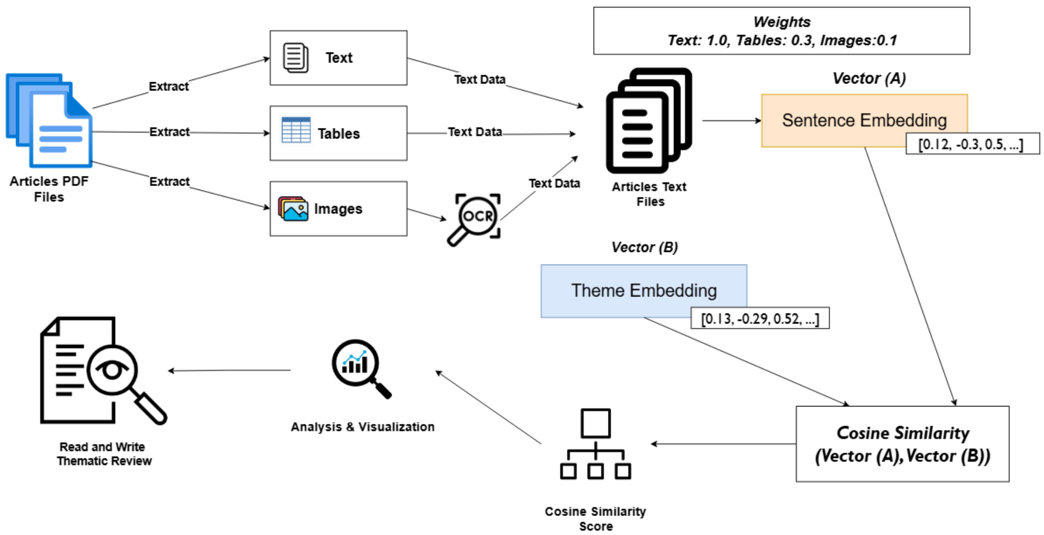

Figure 3: Semantic Filtering and Scoring Methodology

The workflow for this stage is formalised in two algorithms. `Algorithm` 1. crawls through all PDFs in the collection and extract texts as mentioned and pushing all of that together into one structured text file per article. `Algorithm` 2. process those files line by line, and encodes each of those lines in *all-MiniLM-L6-v2* as stated previously, and then the cosine similarity with an array of reference vectors containing our review themes and keywords is later computed. The latter similarity scores yield automatic rankings of each paper relative to their thematic relevance with an anticipated repeatable format that will be followed subsequently by reviews in each theme.

Algorithm 1: PDF-to-Text

Input: $P = \{p_1, p_2, \ldots, p_n\}$: Set of PDFs
Output: $T = \{t_1, t_2, \ldots, t_n\}$: Set of structured text files

foreach $p_i \in P$ **do**
  Text Extraction:;
  $M_i \leftarrow \text{PDFMiner}(p_i)$;
  if $M_i = \emptyset$ **then**
    $M_i \leftarrow \text{OCR}(\text{convert\_to\_images}(p_i))$;
  Table Extraction:;
  $B_i \leftarrow \text{extract\_tables}(p_i)$;
  Image Extraction:;
  $I_i \leftarrow \text{extract\_images}(p_i)$;
  OCR on Images:;
  $O_i \leftarrow \sum_{\text{img} \in I_i} \text{OCR}(\text{preprocess}(\text{img}))$;
  Cleaning:;
  $\text{clean}(M_i, B_i, O_i)$;
  Combine into Structured Text:;
  $t_i = [\text{TEXT: } M_i; \text{TABLES: } B_i; \text{IMAGES: } I_i; \text{OCR TEXT: } O_i]$;
return $T = \{t_1, \ldots, t_n\}$;

`Algorithm` 1 accepts a set of PDF files $P = \{p_1, p_2, \ldots, p_n\}$ and produces a corresponding set of structured text files $T = \{t_1, t_2, \ldots, t_n\}$. To do it with each document p_i, the process starts with extracting the primary textual data using `PDFMiner`. In the case of a file that does not include an embedded text layer (e.g. scanned text documents), an optical character recognition (OCR) solution is performed after conversion scanned the pages as pictures. Afterward, tables are retrieved using `pdfplumber`, and embedded images are extracted via `PyMuPDF`. OCR is then performed on these images after basic preprocessing (grayscale conversion and thresholding) to recover any text embedded within figures. The extracted components; main text M_i, tables B_i, image references I_i, and OCR text O_i; are cleaned to remove noise and merged into a structured output file containing four labelled sections: **TEXT**, **TABLES**, **IMAGES**, and **OCR TEXT**. This final text representation serves as the foundation for downstream semantic analysis and ranking.

Algorithm 2: Semantic Ranking to Themes

Input: $D = \{d_1, d_2, \ldots, d_n\}$: Structured text files from Algorithm 1; $T = \{\tau_1, \tau_2, \ldots, \tau_m\}$: Predefined theme descriptions; ω: Weight vector for sections (*text*=1.0, *tables*=0.3, *images*=0.2); Thresholds: $\theta_{direct} = 0.70$, $\theta_{partial} = 0.50$.

Output: Matrix $R \in \mathbb{R}^{n \times m}$: similarity scores; annotated CSV with relevance labels.

Embed Themes:;
foreach $\tau_j \in T$ **do**
  $E_{\tau_j} \leftarrow$ SentenceTransformer(τ_j);

foreach $d_i \in D$ **do**
  Partition d_i **into sections:** $\{s_{text}, s_{tables}, s_{images}\}$;
  Split sections into sentences where $|u| \geq 8$ words;
  Initialise $max\_score[\tau_j] = 0$ and $top\_sentence[\tau_j] = \emptyset$ for all τ_j;
  foreach *section* s_k *with weight* ω_k **do**
    foreach *sentence* $u \in s_k$ **do**
      $e_u \leftarrow$ SentenceTransformer(u);
      foreach $\tau_j \in T$ **do**
        $sim \leftarrow \text{cosine}(e_u, E_{\tau_j}) \times \omega_k$;
        if $sim > max\_score[\tau_j]$ **then**
          $max\_score[\tau_j] \leftarrow sim$;
          $top\_sentence[\tau_j] \leftarrow u$;

  Assign Relevance Label for each τ_j**:**;
  if $max\_score[\tau_j] \geq \theta_{direct}$ **then**
    label $\leftarrow$ "direct";
  else if $\theta_{partial} \leq max\_score[\tau_j] < \theta_{direct}$ **then**
    label $\leftarrow$ "partial";
  else
    label $\leftarrow$ "not addressed";
  Store results for d_i in R with labels and *top_sentence*;

Export R;

To evaluate each paper against the key directions of current ACC research, the conceptual framwork themes serve as target vectors in the semantic-ranking procedure detailed in `Algorithm` 2.

`Algorithm` 2 takes as input the corpus of structured text files $T = \{t_1, t_2, \ldots, t_n\}$ produced in `Algorithm` 1 and returns, for every article t_i, a vector of thematic labels $L_i = \langle \ell_i^1, \ell_i^2, \ldots, \ell_i^{10} \rangle$ together with the associated similarity scores $S_i = \langle \sigma_i^1, \sigma_i^2, \ldots, \sigma_i^{10} \rangle$.

To initialise the procedure, each research theme mentioned above θ_j ($j = 1, \ldots, 10$) is represented by a reference embedding $e(\theta_j)$ obtained from its narrative description and keyword list (See Appendix A). The 384-dimensional vectors constitute the anchor set $E = \{e(\theta_1), e(\theta_2), \ldots, e(\theta_{10})\}$.

For every document $t_i \in T$, the algorithm first partitions the content into three labelled sections: main body M_i, tables B_i, and OCR-derived figure text O_i. Each section is tokenised into sentences, and sentences shorter than a predefined length are discarded. Every remaining sentence s is embedded to obtain $e(s)$. Cosine similarity is then computed between $e(s)$ and each theme vector $e(\theta_j)$. The resulting score is multiplied by a section-specific weight $w(M) = 1.0$, $w(B) = 0.3$, and $w(O) = 0.2$ to reflect the relative semantic richness of the source. For each theme θ_j, the algorithm retains the maximum weighted similarity.

$$\sigma_{ij} = \max_{s \in \{M_i, B_i, O_i\}} \big[w(\text{section}(s)) \cdot \cos(e(s), e(\theta_j)) \big],$$

and the algorithm stores the sentence s that achieves this maximum.
The final labelling step assigns

$$\ell_{ij} = \begin{cases} \text{“directly addressed”}, & \sigma_{ij} \geq 0.70, \\ \text{“partially addressed”}, & 0.50 \leq \sigma_{ij} < 0.70, \\ \text{“not addressed”}, & \sigma_{ij} < 0.50. \end{cases}$$

producing the thematic profile (L_i, S_i) for article t_i. Collecting these outputs for all documents yields a thematic similarity matrix that maps the corpus onto the ten research themes and provides the evidence sentences that justify each assignment, thereby supporting subsequent quantitative synthesis and qualitative review.

2.6. Findings and Pre-Thematic Synthesis

Before proceeding to the entire thematic synthesis, we should demonstrate how the ranking matrix used to drive our analysis. This matrix cross-tabulates 89 peer-reviewed papers with ten conceptual themes that together conceptualise ACC after 10 review papers were excluded. Table 3 shows a sample version of the matrix, showing only similarity scores of five exemplar themes. The entire matrix, in addition and all supporting project files, is found in the online `GitHub` repository (short-url-link). This section is presented with the sole purpose to illustrate the scoring scheme and thematic labelling can be found in the full matrix; the complete observation of all ten themes is provided in the following sections.

Paper Title	**Multimodal Information Extraction**	**Formalisation of Regulatory Text**	**Semantic Alignment with BIM/IFC**	**Integration of Ontologies and Knowledge Graphs**	**Rule Representation and Reasoning**
Hierarchical Representation and DL-Based Methods...	0.64	0.64	0.67	0.67	0.61
Knowledge graph exploitation to enhance...	0.50	0.58	0.68	0.51	0.50
Converting Fire Safety Regulations to SHACL...	0.61	0.67	0.66	0.64	0.68
Automated Building Information Modeling...	0.70	0.73	0.71	0.67	0.72
Integrating NLP and context free grammar...	0.66	0.72	0.64	0.68	0.70

Table 3: Sample from the Thematic Similarity Matrix

2.6.1. Depth Versus Breadth in Thematic Engagement

Figure 4 denotes an asymmetry of depth of engagement. Despite the prevalence of Rule Representation and Reasoning and Formalisation of Regulatory Text (90 and 75 papers, respectively), only one-third of the entries can be considered as directly engaged, the other ones are labelled as partial. Integration of Ontologies and Knowledge Graphs, in turn, considering significantly fewer contributions (25 papers) has the greatest direct to partial ratio, which means that the area is smaller, but more methodologically sound. At the other themes such as Human-in-the-Loop Approaches and Explainability and Trust in AI Systems have extremely low direct coverage, and in fact, are continually sidelined, despite their strategic value to user-centred and trustworthy ACC systems.

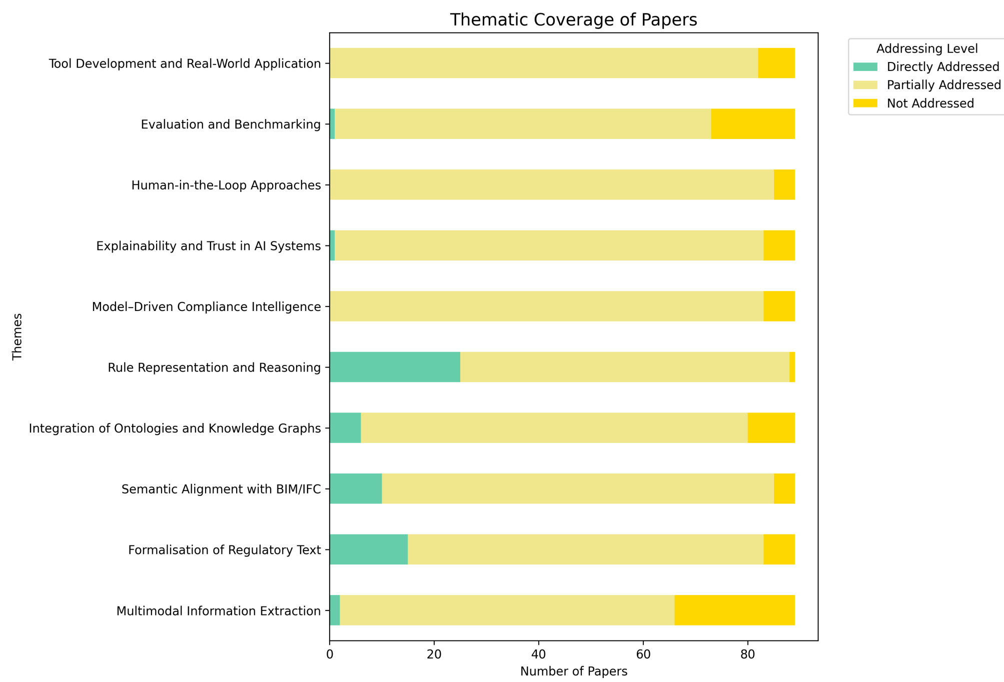

Figure 4: Stacked Horizontal Bar Chart

2.6.2. Structural Segmentation Revealed by the Heatmap

Figure 5 shows two latent research clusters that are aligned on methodological fault lines. The upper-left cluster is the combination of the papers with more than 0.70 scores in Semantic Alignment with BIM/IFC and Ontology Integration in the spirit of knowledge-engineering orientation. Conversely, the papers in the lower-right cluster focus on Tool Development and Real-World Application coupled with Human-in-the-Loop Approaches, and this implies an implementation-oriented attitude. The white space between these groups of Explainability and Evaluation highlights the near absence of integrative work which brings together the upstream sophistication of modelling with the downstream validation and usability. The gap is one of the bottlenecks to end-to-end ACC workflows.

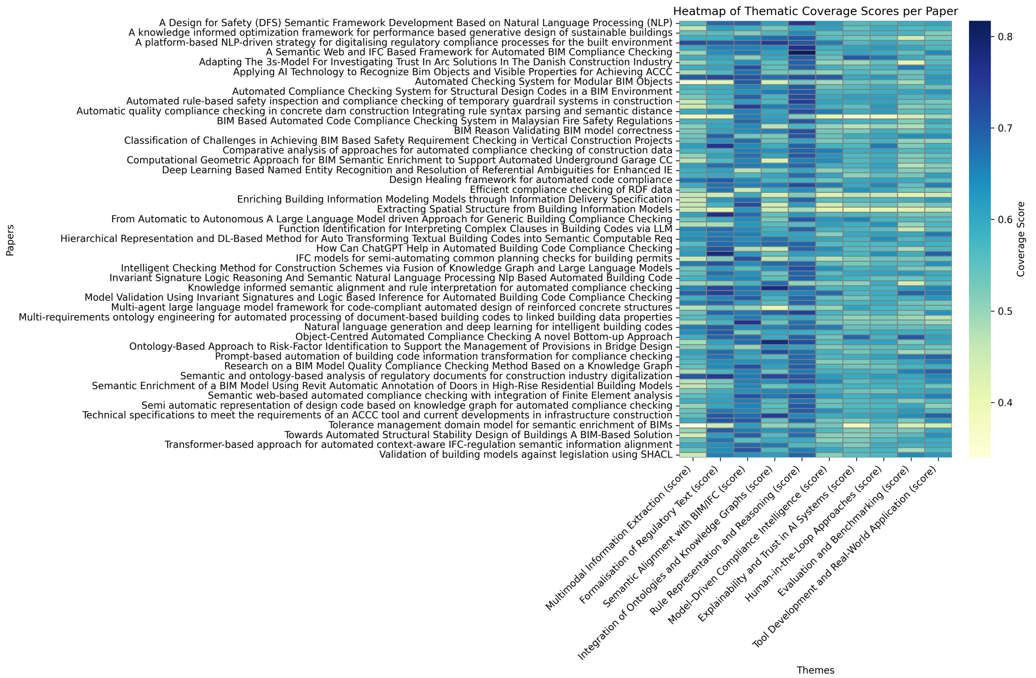

Figure 5: Paper-Level Thematic Similarity Matrix

2.6.3. Peaks, Plateaus and Performance Bottleneck

Figure 6 reveals that the difference among the themes is limited by the rule representation and reasoning which has the highest score 0.66 and integration of ontologies being its immediate neighbour. In these tops, one finds a midfield level: six themes have been clustered between 0.56 and 0.60, demonstrating homogenous incrementalism instead of the differentiated progress. Such plateau is the expression of the fact that it is dependent on other infrastructures (like a shared reliance on IFC schemas or SHACL validators) and can only diversify their themes to a limited degree. Multimodal Information Extraction has not seen a large scale growth at the bottom (0.54), and Tool Development and Real-World Application have not grown much at all and skewed, which implies that the interest is beginning to rise, though not in any coordinated manner.

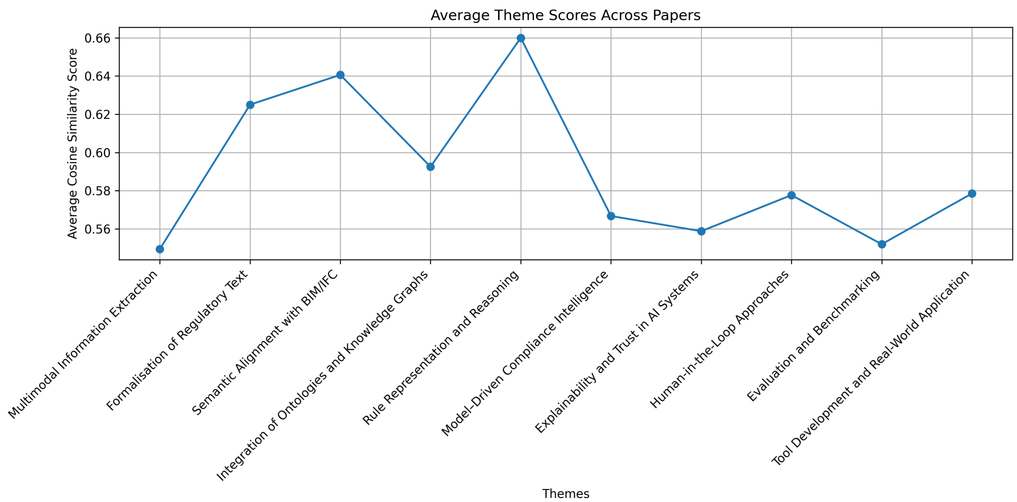

Figure 6: Average Thematic Scores

2.6.4. Structural Fragmentation in Paper-Theme Flows

The Sankey diagram shown in Figure 7 has directional asymmetry and structural isolation in the forefront. The eight themes from ten represent all the significant affiliations, and Evaluation and Benchmarking and Explainability are the two near leaves, and they rarely act as a major anchor. There are parallel currents of thickness toward Semantic Alignment and Ontology Integration that shows that there is minimal branching, and this is evidence that innovation flows do not intersect, only parallel. This linearity restricts systemic integration, which justifies a model of technical depth creation that is not converted into holistic solutions to compliance. Research agendas will need to overcome such structural silos by integrating representational rigour with interpretability, validation, and deployment.

2.6.5. Structural Fragmentation in Paper-Theme Flows

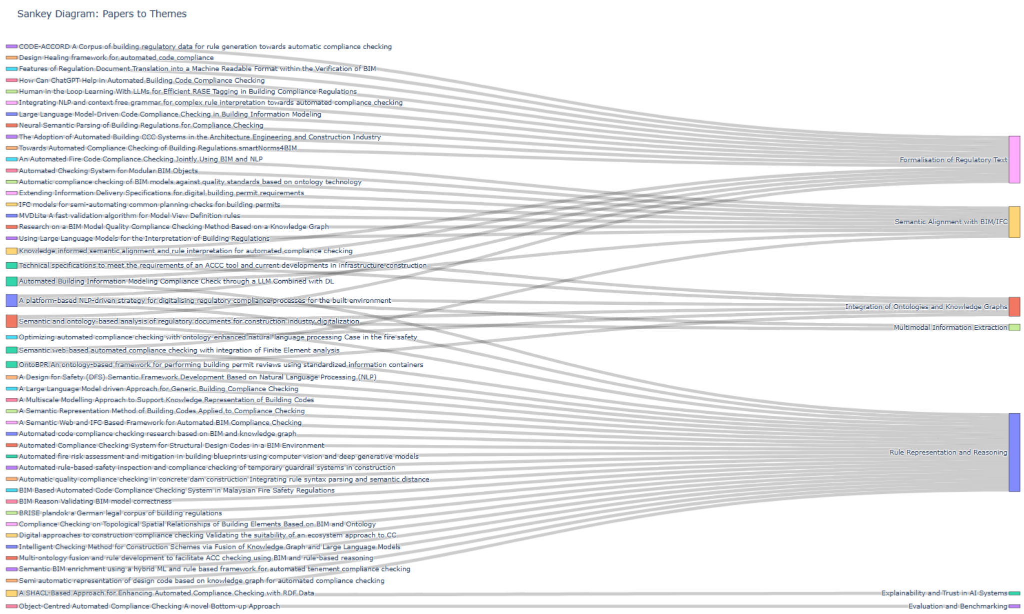

Figure 7: Sankey Diagram

2.6.6. Ceiling Effects and Thematic Development Profiles

Figure 8 shows the difference in the thematic development profiles. The upstream domains (Formalisation and Rule Representation) show that score bands (0.78-0.82 of their top five papers) are compressed, which means that approaches are converging, and methods are stable in terms of evaluation. In the sharp contrast, Explainability and Human-in-the-Loop Approaches exhibit low ceilings (0.70) and more dispersion, which is typical of conceptual volatility and fragmented evaluation criteria. These disparities characterize the thematic core margin structure of the ACC study and the location of consolidation that has been challenging.

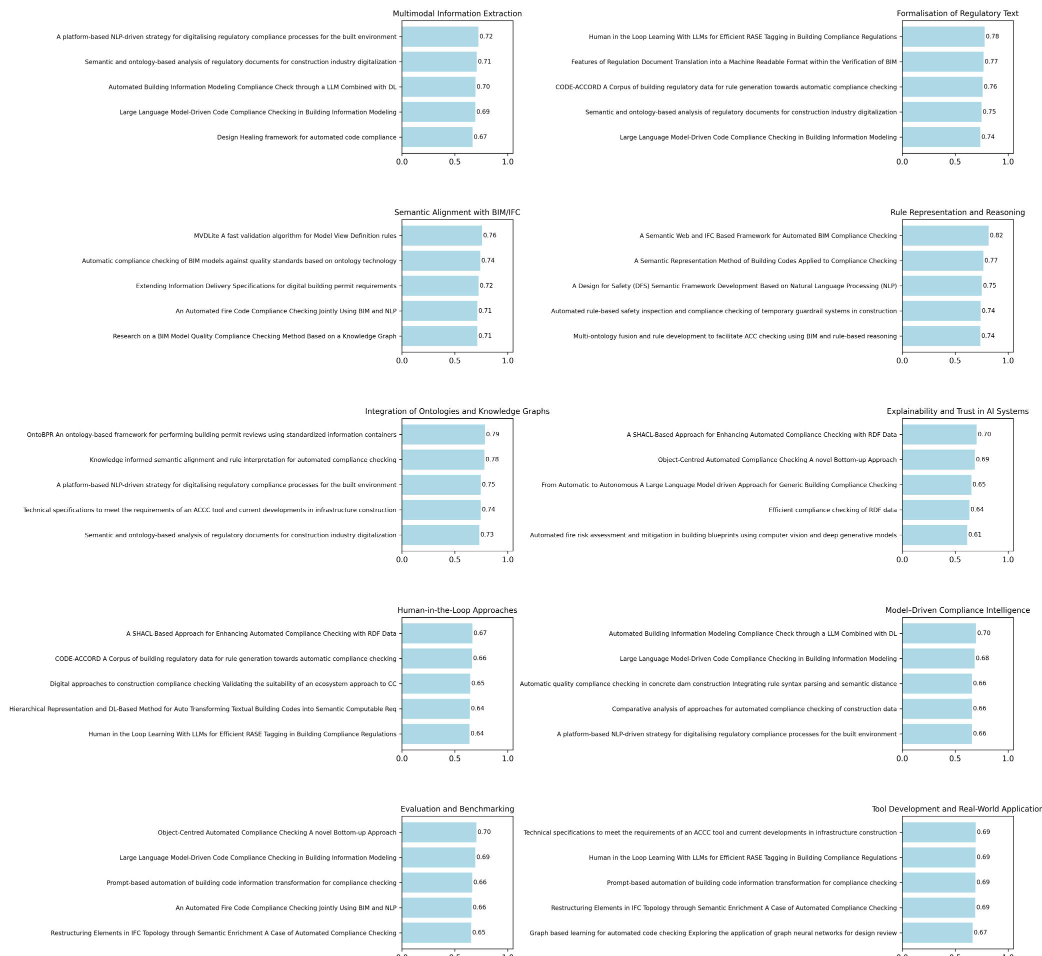

Figure 8: Top-Five Ranked Papers Per Theme

3. Thematic Synthesis

To support this thematic synthesis, we constructed a summary matrix of the selected literature, comprising the top-ranked papers drawn from 89 peer-reviewed studies. Notable attributes provided in this matrix are the names of the authors, goal, method, tools, domain, contribution, mapping of themes, and the binary indicator of the theme relevance (Yes/No) (See Appendix B).

The study started with a systematic filtering and close reading of 33 papers. Three of these did not fit with the initially scoring themes [38, 41, 42]. Rather than excluding them, we examined them thematically and updated manually with the help of domain experts to make them consistent and relevant.

The following subsections examine the ten identified themes. In every theme, top ranking papers in the themes are critically examined. In each of these papers we give a clear summary and a sharp evaluation, including citation, method, tools. The conclusion of each theme points to gaps in knowledge and emergent issues as well as future research directions. In this way, we can offer a combination of recent developments that are informative and offer a clear perspective of where the ACC research has reached and the possibilities that lay ahead.

3.1. Multimodal Information Extraction

Multimodal Information Extraction (MIE) for regulatory compliance aims to turn heterogeneous artefacts such as legal prose, tables, and drawings into computable structures. N. Chen et al. implement a two-stage textual pipeline. In the first stage they pre-classify clauses using Text Convolutional Neural Network (T-CNN), Long short-term memory (LSTM), Bidirectional Encoder Representations from Transformers (BERT), and a fusion model. In the second stage a large language model performs structured extraction, which is mapped to an ontology and executed with SPARQL [37]. A complementary approach, Madireddy et al. treat formalisation as direct code synthesis, using large language models to interpret textual specifications alongside BIM parameters and to emit executable checks [51].

For visual inputs, D. Chen et al. present a vision-only workflow processes scanned or digital floor plans. Images are standardised, contours and lines are detected, colour-coded symbols and on-drawing text are recognised with OpenCV and Tesseract OCR, and two outputs are produced: a vectorised floor plan and blueprint metadata (room types, dimensions, materials, fire-safety facilities). The workflow does not require BIM/IFC, enabling automated checking from images alone [38].

At corpus scale, Kruiper et al. propose the I-REC platform processes full regulatory collections using a shallow sequence tagger (SPAR.txt) to detect multi-word expressions and build a domain lexicon/knowledge graph; they highlight difficulties in PDF layouts (lists, figures, document structure) and the need for layout-aware parsing [31]. Kabzhan et al. propose an ontology-anchored pipeline applies tokenisation, part-of-speech tagging, dependency parsing, and co-reference resolution to produce regulatory statement profiles with fields for subject, predicate, object, modality, and conditions, and identifies automatic extraction of document structure as important future work [58].

MIE is converging around a modality-free pipeline in which heterogeneous regulatory artefacts such as text, drawings and corpus-scale collections are transformed into a shared, machine actionable substrate of ontology-aligned triples, executable constraints, and structured geometry. Regardless of entry point (text-first, code synthesis, vision or even corpus approaches), the common goal is to ground obligations in explicit objects, relations, and parameters; to provide interfaces for querying and executing these obligations across tools and datasets; and to enable reasoning that is reliable, auditable, and reusable.

3.2. Formalisation of Regulatory Text

Formalisation represents an important stage of the compliance pipeline. It turns normative prose into computable artefacts such as annotated corpora, logic encodings, ontology-aligned triples, and executable rules. Hettiarachchi et al. provide the data layer by manually annotating sentence-level excerpts with entities and relations, creating ground truth for NER, relation extraction, and sequence-to-sequence rule generation; the authors motivate supervised and generative models such as T5 or BART for mapping to Legal Rule Markup Language (LegalRuleML) and promote annotations that generalise beyond a single domain ontolog [14].

Ren et al. clean specification text, tags key entities with a BERT, A Bidirectional LSTM (BiLSTM), Conditional random field (CRF) model that improves over a BiLSTM baseline, parses tagged spans into syntax trees with ANother Tool for Language Recognition (ANTLR), and align rule and record trees using Simbert Chinese Tiny LLM (SimBERT) with numeric phrases normalised to operators [41].

Makisha proposes a two-stage RuleML pipeline. The first stage, provisions are decomposed into First-Order logical atoms (predicates, variables, quantifiers). The second stage, they are rendered in Rule Markup Language (RuleML) while aligning terms with IFC vocabulary. The method also classifies clauses that resist automation (cross-references, ambiguous statements, non-verifiable on BIM) [46].

A complementary step formalizes via Requirement, Applicability, Selection, Exception (RASE). Al-Turki et al. convert Regulatory prose into a structured Ain't Markup Language (YAML) that preserves document hierarchy, encodes RASE tags with fine-grained metadata. Non-textual content is stored as base64-encoded images and tagged tables, and subsequently workflows align or translate Building Compliance Rule Language (BCRL) to a logic-ontology layer [49].

In parallel, Madireddy et al. treat formalisation as direct synthesis of executable checks. An LLM emits Revit-ready Python, with a feedback loop that re-prompts on runtime errors until scripts execute and produce compliance reports effectively, but it is a tool dependent formalization [51]. Finally, Kabzhan et al. use Simple Knowledge Organization System (SKOS), which is a W3C standard Resource Description Framework (RDF) to build regulatory statement profiles for scalable integration in ACC workflows [58].

Formalisation is the primary axis between regulatory prose and automated checking. There are two convergent trends across approaches towards a common field; a set of canonical representations (RuleML/LegalRuleML, RDF/SKOS, and RASE structures) and directly executable artefacts. These two are tethered to IFC semantics. The resulting shared substrate is an agreed-upon, machine actionable substrate that retains document hierarchy, meaning and allows programmatic evaluation in the form of queries or code and can be tracked backward to the original text. This path is making compliance logic more interoperable, audit able and reusable, across projects and jurisdictions.

3.3. Semantic Alignment with BIM/IFC

Semantic alignment with BIM/IFC forms a critical stage in the compliance pipeline, which is achieved by binding regulatory concepts and rule predicates to the IFC data model so that checks can be evaluated over classes, attributes, and relations defined by the standard. Wang et al. achieve semantics with BIM by treating IFC as the authoritative data model and compiling code clauses into logical predicates evaluated over IFC attributes and relations. They discuss alternative semantic encodings of IFC via ifcOWL/RDF with SPARQL while ultimately extracting IFC entities/attributes and augmenting them with a geometry-processing layer to resolve spatial relationships required by fire-code rules. Compliance is then determined by executing these expressions on IFC-native data and the derived spatial relationships [36].

Fischer et al. extend buildingSMART's Information Delivery Specification (IDS) to express information requirements that are explicitly tied to

IFC semantics. IDS files specify the information that an IFC ought to have. Recent extensions allow specifications to filter the elements based on related elements in IFC, for example selecting those doors belonging to a particular storey or those between a dwelling and a corridor. These additions are followed by escape route tracking, fire resistant verification and develop directly on the IDS aspects connected to IFC entities, attributes, relations, and properties [45].

Liu et al. pursue semantic alignment by binding Information Delivery Manual (IDM) requirements to IFC through Model View Definition XML (MVDXML). Each domain requirement is encoded as an "information unit" that compiles into a subgraph pattern over the IFC instance graph (a DAG of entities, attributes, and typed relations) together with declarative value constraints. Validation then proceeds by matching these templates to the IFC graph and evaluating the associated constraints on the matched nodes. Their contribution accelerates traversal and matching while preserving the established semantics of MVDXML [52].

Ma et al. adopt a different strategy, encoding the BIM quality standard as an OWL domain ontology and expressing the verification logic as SWRL rules. They then populate this ontology with data drawn from a targeted subset of the IFC model by mapping ontology classes and properties to the corresponding IFC entities and attributes. This allows a rule engine to reason over IFC-derived individuals and infer compliance directly. Although an ifcOWL/RDF/SPARQL pathway is known, the authors adopt a bespoke OWL vocabulary tightly coupled to IFC and execute SWRL rules and visualizing results in BIMServer. The outcome is a traceable, explainable bridge from IFC data to normative concepts that supports automated, ontology-driven compliance checking [40].

These reviewed studies align regulatory concepts to BIM/IFC in the form of: SWRL/OWL rules over instances of IFC, IDS information specifications, MVDXML validation of IFC graphs and ifcOWL/SPARQL queries. IDS largely focuses with alphanumeric characteristics; they are based on user choice and verification geometry by relation and topology. Large corpora may be resource-intensive for MVDXML validation. Ontology-IFC mappings should always be updated irrespective of which approaches should be used since the models and schemes will always be modified. Text-to-IFC alignment would likely benefit from stronger cross-reference handling, more expressive related-element filters, and shared libraries of mapping support would also benefit every day AEC practice.

3.4. Integration of Ontologies and Knowledge Graphs (KGs)

At this stage, ontologies and KGs move beyond being part of the supporting metadata to constitute core infrastructure that glues regulatory text, BIM model, and the review process together. Fauth et al. in OntoBPR model the building permit process as a layered ontology, which integrated a variety of domain ontologies, not through one, monolithic model, but through a malleable upper model. Then they encoded documents, data, and executable rules in Information Container for linked Document Delivery (ICDD) containers as RDF, and validated them with SHACL, and used SPARQL to make decision and notifications. The composition of modular ontologies and binding of these to standardized containers and the imposition of graph-native constraints are the means of achieving integration [54].

Zheng et al. construct a domain knowledge graph in which encodes regulatory, BIM concepts, lexical variants, relations, and constraints were modelled using typed nodes and edges. The entities extracted out of regulatory text are matched to this graph in an unsupervised similarity with conflicts disambiguated before use. The resulting KG is used as data and schema backbone to generate and execute SPARQL rules that allows the checker to infer implicit quantities and relationships to the graph structure during validation [50].

From a platform perspective, Kruiper et al. in I-REC platform treat the KG as shared infrastructure. It also automatically generates KGs to facilitate semantic search over a large body of regulations and then it can leverage the KG to extend queries and documents in such a way that terminological variation does not interfere with retrieval. The platform also recognized the governance problem of the corpus tools that can determine the new concepts, update the KG, and explicitly require domain vocabularies that are aligned and can be united in a shared graph with the guidelines of integrating project-specific terminologies. In this case, integration is perceived as a socio-technical where the KG is considered as a technical artifact and a vocabulary-alignment coordination device [31].

In Kabzhan et al., the knowledge graph is implemented as an RDF triple store (Apache Jena Fuseki) which is hosting a layered ontology stack that comprised Basic Formal Ontology (BFO)-based upper with IFC-aligned domain, deontic model of regulatory statements, and SKOS terminology layer that is accessible by SPARQL query with SHACL validation via the Datavera Enterprise Knowledge Graph (EKG) provider. Text fragments are projected to this schema and encoded as semantic profiles as stated previously, filling the KG to be used in downstream processing, and terms are related to concepts using SKOS with ad-hoc concepts being formed when no perfect match is found. Graph queries over these profiles then support cross-document detection of duplicated provisions and contradictions directly in the KG [58].

Schuk et al. integrate ontology and graphs database to deal with infrastructure rules. They model regulations, rule nodes and infrastructure objects in a specific node-edge pattern, rule faceted classification and apply forward and backward rule-to-rule chaining to express dependencies. The graph can be used to ensure relations, rules hierarchies can be observed and traced to ensure automatic checking of code compliance in transport works [59].

Throughout the recent researches, layered ontologies were aligned to enable work run checks on RDF data in ICDD containers with SHACL and SPARQL, text to ontology pipelines transform rule text into SPARQL to BIM checking, living knowledge graphs connect SKOS terms, regulatory statements and IFC concepts in order to allow alignment of vocabularies, and graph databases use explicit node and edge patterns with rule and object aspects as well as forward and backward chaining to make rule hierarchies understandable and traceable, while the key gaps is the shared vocabulary alignment and licensing between parties, sustained maintenance of graphs as rules evolve, effective governance of rules authoring and trackability back to the source text, and the absence of a standard schema in the case of infrastructure rules, so that treating ontologies and knowledge graphs as core AEC infrastructure makes integration an ongoing capability instead of a one off mapping.

3.5. Rule Representation and Reasoning

The compliance stage of the conceptual ACC framework that allows the transformation of regulatory requirements into the representation that can be evaluated by computational systems is the rule representation and reasoning. Across recent AEC studies, rule languages provide the semantic foundation of the representation and reasoning of regulatory requirements.

Zhou et al., Li et al., and Jiang et al. all involves a mapping between regulatory concepts and BIM or IFC entities and properties so that compliance checks can be performed by ontology-based inference. Zhou et al. and Li et al. introduce rules in SWRL over OWL ontologies; Li also uses an intermediate constraint schema that is automatically transformed to SWRL, which enables the description of relational and spatial constraints [30, 33].

Jiang et al. combine a regulation ontology and a design ontology and preserve an independent layer of rules, permitting clear checking rules to be run with transparent, explainable results. Such a family of approaches is more concerned with semantic validation, logical rigour, and traceability between the encoded rule and its source concept [1].

Jia et al. present rules as SPARQL query templates on top of ontologies in ifcOWL in which the query itself is taken to be the executable constraint. The benefits of these techniques are the declarative representations, graphical reasoning, and the compatibility with the current semantic web infrastructure [34]. Finally, Johansen et al. make the requirements of guardrail inspection explicit and parametrised as rules and applies them to captured attributes of sites or models. Although less formally expressive, these approaches focus on flexibility, compatibility with the current workflows and simple applicability to model data [39].

All these studies follow a similar pattern overall: express the rules in a formal language (SWRL, SPARQL templates, ontology axioms), relate to BIM/IFC through explicit mappings, and reason with the help of reasoning engines to yield explainable output. The key gaps remain evident in the first three stages Multimodal Information Extraction, Formalisation of Regulatory Text, and Rule Representation and Reasoning are: layout- and structure-aware parsing of complex PDFs (lists, tables, figures, headings) with explicit modelling of cross-references and document hierarchy; cross-modal alignment that grounds each textual requirement in a specific spatial/semantic entity in drawings or BIM; faithful handling of nested conditions, exceptions, and deontic modifiers in an executable and traceable form; and robust, durable mappings between regulatory terms and IFC/ontologies despite schema evolution and gaps in geometric/topological coverage.

3.6. Model–Driven Compliance Intelligence

LLMs are integrated in two complementary phases: supporting information extraction and subsequent rule formalization to enhance efficiency across both stages. Recent studies deploy Large Language Models (LLMs) to interpret regulatory text into computable representations and augment domain knowledge, so those representations are usable in downstream checking.

A hybrid pipeline mentioned by N. Chen et al. combines an GPT-4 with deep-learning pre-classification and a domain ontology: a classifier first processes regulatory passages; an GPT-4 is then able to implement structured information extraction; and a construction ontology provides a set of concepts with which to stabilize interpretation of nested and conditional statements. Its design can leverage LLM few-shot capabilities to reduce annotation requirements and increase faithfulness of extraction. This setup focuses on quantitative checks such as areas and heights and reports plans to use instruction fine-tuning or LoRA to reduce the pre-classification step [37].

Madireddy et al. they use ChatGPT, Claude, Gemini, and LLaMA to understand codes, produce Python scripts. To illustrate how LLMs play the role of knowledge amplifiers where they become part of design tools, case studies report fewer manual tasks, reduced amounts of known violation-discovery time [51].

D. Chen et al. and Kruiper et al. provide complementary enablers for LLM-based compliance in AEC. A computer vision pipeline turns plans into structured facts and suggested redesigns, which gives the LLM concrete inputs to reason about. A retrieval layer like i-REC brings the right clauses and definitions, which grounds the LLM's answers in evidence. Joined together, the LLM can cite rules, explain non-compliance in plain language, compare redesigns to requirements, and produce audit-ready outputs [31, 38]. Moreover, Nuyts et al. compare eight rule technologies and find SHACL the most suitable for validation in construction data, which makes it a good target for the output of LLM pipelines and for the graph layer that executes checks at scale [6].

LLMs are portrayed as a facilitating step towards compliance. They can translate regulatory prose into executable rules, introduce domain knowledge into extraction, and enabling integrated with BIM systems. They will likewise consume less data and yield more robust outcomes when used together with prompting, pre-classification, ontologies, and human oversight can cut data requirements and improve reliability. Key gaps remain: few instruction-tuned LLMs for AEC rules, tool lock-in and variable outputs across models, limits in licensing and vocabulary coverage for retrieval, and weak generalization beyond geometry and topology. A practical stack is to let retrieval and knowledge graphs feed the prompts, have LLMs extract or generate checks, run logic with SHACL or SPARQL, and use computer vision when BIM is missing.

3.7. Explainability and Trust in AI Systems

Explainability comes from making the reasoning clear and open to inspection, either as symbolic rules on a knowledge graph or as plain-language narratives, so reviewers can see why each decision was made. Robaldo et al.

argue that trust in legal and AEC compliance tools grows when decisions come from symbolic rules rather than opaque models, since machine learning often acts like a black box and is hard to justify. They show how using RDF with SHACL, OWL, and SPARQL helps produce human-readable reasoning chains, which gives clear explanations and supports audit work. Their view is that explainable AI in law and compliance should rest on standard web technologies so that every inference can be traced and checked [44]. Anim et al. push this idea further by encoding rule logic as SHACL-SPARQL over RDF so that each pass or fail is tied to explicit constraints and to the data that triggered them. They motivate symbolic rules as a way to show the steps of reasoning, which makes outcomes interpretable to reviewers and lawyers. They also discuss adding large language models to extract inputs, while keeping the final decision path anchored in rules for clarity and audit [35].

Ying and Sacks design an LLM-driven checker for buildings that aims for transparency by writing every check and result in plain language, which helps teams understand why a rule passed or failed. They plan richer reporting, including natural language logs and 3D views of results, so designers can audit decisions inside design tools. They also warn about risks that matter for trust, such as variable answers, the need for robust testing, and data privacy when models handle project files. [32].

Doukari et al. illustrate a Bottom up, object-centred checker which notifies every basic test with the reason for non-compliance when it fails. Their method puts checks on simple objects to complex objects and on mandatory rules to guidance and then provides a comprehensive report that can be followed by the users. This gradual report is more reassuring since the reviewers are in a position to notice what was tested and in which sequence and the reason why it has failed [53].

Across these AEC studies, explainability comes from two paths: using symbolic rules on knowledge graphs that leave a clear audit trail, and providing natural language reports that show what was checked and why inside the design workflow. The main gaps are the heavy manual work to build and maintain rule bases, variability and privacy risks in large language models that demand guardrails and evaluation, the limited scalability of hand written if then rules, and the need for reliable pipelines that combine LLM extraction with rule based decisions anchored to graphs. The shared goal is to link data extraction to symbolic rules with stable reporting and provenance so every compliance result is understandable and auditable.

3.8. Human-in-the-Loop Approaches

A complementary stage with the expendability is to involve human expertise within the pipeline, not as an afterthought, but as a supervisory mechanism that manage models and validates outcomes. Anim et al. place people at the centre of rule building and validation. They ask legal and domain experts to craft and review the ontology and the SHACL SPARQL rules that run on RDF data, and they use structured methods like MELON and the O2 framework to guide that expert work. Their aim is to keep experts in the loop for writing, debugging, and auditing rules, while planning later support from large language models to ease the manual load [35].

Al Turki et al. design a human in the loop pipeline for RASE tagging of building rules. They drive an LLM with few shot and fine tuning setups and then use progressive active learning, where experts review the model output and feed back corrections in cycles. The team records both structure and meaning in YAML so that later rule generation stays traceable to expert edits [49].

Zhang and El Gohary show where people still matter even with deep learning. They propose a method that learns requirement hierarchies from text, but they note that current rule encodings and many prior systems depend on professionals who read codes, mark key roles, and check results. Their work reduces manual steps, yet still assumes labelled data and expert validation to keep quality high [48]. After that, Hettiarachchi et al. build a strong human loop. Twelve annotators label entities and relations in sentences from English and Finnish building rules, and the authors then curate the labels. This creates the ground truth that later models learn from, and it also exposes where experts agree or disagree about rule meaning [14].

Beach et al. argue for an ecosystem where human roles are explicit across services. They describe workflows in which regulation experts mark up rules with RASE, BIM experts map rules to model data, and reviewers read results in interfaces that are made for understanding. Their validation of a distributed architecture underlines that people supply key inputs and approvals at several points, not just at the end [43].

Key gaps in human-in-the-loop compliance for AEC include slow and costly creation and update of rules and ontologies, limited ways to share editing effort and track quality, and weak review workflows without clear protocols or error dashboards. Labeled data is expensive and often too narrow to cover the many code styles in practice, while learned structures still need expert vetting before use in permits. Tools are also siloed, making it hard for experts to add judgments across platforms. These issues point to a common need: standard roles, structured review stages, open interfaces, shared data schemas, and full audit trails so human oversight is consistent, measurable, and easy to plug into project systems.

3.9. Evaluation and Benchmarking

The final stage is evaluation and benchmarking showing how recent ACC studies validate their work: what they accept as ground truth, which metrics they use, and how they run head-to-head comparisons.

Several works adopt expert-reviewed gold standards and report classic IR metrics. Wang et al. test 10 IFC models of various building types, compare system outputs with the results of an expert manual review, and calculate precision, recall, and F-measure per project and averaged over all projects; after ten repeated runs they report mean recall ≈97% (reaching 100% on smaller models), with precision ≈91% using the expert as the gold standard. They describe the evaluation table (gold-standard counts, system detections, true positives), and the formulas of the metrics with the recall being the main criteria of safety-critical checks [36].

Prompt-based LLM transformation of code provisions is benchmarked at clause level. Yang and Zhang test 51 International Building Codes (IBC-2015) requirements and report ≈ 97% precision and ≈ 96% recall for logic-clause generation using only 2% training data, alongside throughput measures (≈60.8s per clause) and a ≈ 28% time saving versus a rule-based baseline—making time-to-result part of the evaluation [55].

Task and stage specific metrics appear in construction phase quality checking. Ren et al. combine information-extraction and alignment benchmarks (entity F1 up by ≈13% over peer models) with end-to-end decision accuracy (≈89%), acknowledging mixed quantitative/qualitative constraints and moving beyond binary pass/fail to a scored outcome [55].

The timing of case-studies is also applied as a yardstick of comparison. Doukari et al. evaluate a bottom-up, object-centred checker through two case studies that use manufacturer BIM object libraries. They report practical effort gains rather than classification metrics and show savings of about 125 minutes per object when compared with manual workflows. This process-focused measure is relevant to offices that judge tools by throughput and rework avoided. [53].

The LLM-in-BIM pipelines add model-comparison benchmarks. Madireddy et al. compare various LLMs on the time of generating a script, they observe that GPT-4 and Claude succeed where some of the other models fail, which is useful in selecting a method but constrained by the relatively small set of rules [51].

Karmakar and Delhi test semantic enrichment for IFC topology with an Indian development control rule. They demonstrate pass and fail outcomes on a targeted clause and show how enrichment supports automated checks when native IFC relations are missing, which serves as a focused test case template for geometry-heavy rules [57].

AEC benchmarking divides into two valuable paths: formal accuracy trial compared to controlled gold standards of experts with clear metrics, and case trials that estimate time, feasibility or intended rule achievement with BIM processes. Such gaps persist as low cross-domain and cross-region generalization, few multi-jurisdiction test covers, lack of exchanging common public datasets and limited coverage in at-large geometry-topology. The field requires common benchmark corpora, freely accessible BIM models with ground truth, standardised bundle of metrics (precision, recall, F1) and effort (timing, rework) and similarity measures of code to logic translation and repeatable protocols with published rule sets and scripts so that the results become comparable across tools and reviewers.

3.10. Tool Development and Real-World Application

The latest work treats ACC as a software product where architectures are defined, user interface modules are developed, and prototypes are followed in real projects.

Schuk et al. take the next step to a realizable ACC tool design: a dual taxonomy (regulations vs. atomic rules), an interaction model between rules and infrastructure objects, and a graph-database core to follow dense cross-references. They give developer guidance on metadata and where to locate logic and prototype a practitioner-friendly authoring sequence RASE to Business Process Model and Notation (BPMN) and Decision Model and Notation (DMN). The feasibility is demonstrated by the fact that a high percentage of German railway rules (52%) are automatable, and half can be expressed in BPMN/DMN, tested on 943 rules. The graph store is preferred to manage heterogeneity of dependency and inter-standard connections in a more natural way than relational schemas [59].

Yang and Zhang provide two-module system: a frontend executable user-interface to read code clauses and store the generated scripts, and a backend information conversion service to transform in a prompt-driven manner and benchmark it on IBC clauses and in a second domain to demonstrate portability and fewer human efforts. This is proposed as a viable roadmap to productise the code-to-logic conversion in current workflows of the AEC industry [55].

Karmakar et al. exhibit end-to-end usage at project level: they semantically extend IFC to re-discover lost relationships, write a Python checker, and apply it to residential Design Transfer View (IFC4) model. The system produces pass/fail per element keyed on Globally Unique Identifier (GUID) and the measured projections are in the report, and this is an operating pattern that can be examined by authorities and designers [57].

Al-Turki et al. present a rule-authoring frontend where experts turn code text into RASE-structured YAML, correct model outputs, and save machine-readable rules for later use, while Bloch et al. deliver a BIM-to-graph checking pipeline that ingests models, builds labeled graphs, trains a graph neural network, and reports pass or fail in dashboards [49, 47].

ACC is moving from research to practice through end-to-end pipelines that plug into BIM, authoring tools that create structured rules, semantic enrichment that repairs IFC gaps, and platform specifications for infrastructure, forming a deployable chain where rules are authored, models are enriched, checks are executed, and results are reported inside live workflows. Key gaps for real deployment include larger real datasets and stronger baselines, a robust bridge from text outputs to BIM-aware execution, broader validation across rule families and regions, libraries of reusable enrichment patterns, and shared schemas with public test assets for infrastructure so results are consistent across projects and authorities.

4. Discussion

This review indicates that automated compliance checking in AEC is converging around a consistent pipeline: multimodal extraction of regulatory artefacts, their formalisation into machine-readable logic, their semantic alignment to IFC/BIM and representation and reasoning of rules. Throughout these phases, the most common result is a shared machine-actionable layer which various tools can query or execute with the aid of ontology-based rules, SHACL or SPARQL validation and procedural checks that are explicitly bound to model entities. This layer explains handoffs, enables auditability, and minimizes the necessity of end-to-end custom solutions.

According to the thematic synthesis, we observe two dominant clusters in the literature. One is knowledge-engineered focused; this concerns semantic alignment with BIM/IFC and the integration of ontologies and knowledge graphs. The second is implementation oriented, spans tool development in actual projects and human-in-the-loop interactions. The clusters are related to explainability and evaluation, but not anchored by them, which partly explains why fully integrated, auditable end-to-end systems remain quite rare. Innovation often moves in parallel streams, with strong representational work upstream and usability work downstream, and only modest cross pollination between the two.

The degree of maturity also varies in the pipeline. Formalisation and rule representation exhibit tighter method scopes and more stable practice of evaluation, and that means the increase in agreements and replications. Explainability and human oversight is more irregular, and the metrics are heterogeneous, as well as demonstrations at the prototyping level. As a matter of fact, the field is mostly in consensus on how to encode and reason over rules, although it is still refining how to present results to practitioners and how expert judgment should be integrated into typical work processes.

Methodological patterns are powering the strength of current claims. The vast majority of studies justify themselves on constrained corpora or build special models rely on prompts, scripts or ontologies that cannot be publicly reusable. Metrics commonly used such as precision and recall on extraction and pass or fail accuracy on rule execution and do not appear in the literature as consequences of named accuracy measures. There are limited common datasets and gold standard that would allow comparison to be done under control across the areas and jurisdictions.

There are apparent priorities behind these observations. Teams are expected to apply a common semantic core using RDF or OWL with SHACL or SPARQL and create reusable mapping libraries to support semantic alignment, develop outputs executions including a rule identifier, the triggering evidence, linked IFC GUIDs, and a concise rationale. The community ought to publish open benchmarks listing regulatory text coupled with IFC models and gold rule encodings, human in the loop protocols with quantitative targets, and compare end to end pilots of hybrid large language model and symbolic pipelines with purely symbolic and large language model baselines.

5. Conclusion

We set out to clarify the state of automated compliance checking from 2022 to 2025 and to organise a uniformly sampled corpus from Scopus, Web of Science, IEEE Xplore and a deep search with ChatGPT. Using a ten theme framework that spans extraction, formalisation, BIM and IFC alignment, rule representation, ontologies and knowledge graphs, explainability, human in the loop, model driven compliance intelligence, evaluation, and tool deployment, we applied de-duplication, coverage analysis, semantic ranking with MiniLM embeddings, and close reading of the highest ranked papers.

We find our synthesis demonstrates how the field is converging around common semantics. The most typical modes of applying standards to various types of rules in most systems are three common approaches: ontology with SWRL inference, SHACL or SPARQL validation, and procedural checks against model entities. By contrast, integration into production workflows is less mature, and explainability and human supervision receive less systematic attention. Evaluations tend to be expert gold standard based, and report precision, recall, F1 and time to result. These steps are beneficial in terms of comparison but they also indicate the lack of availability of open and reusable test beds.

These observations suggest definite lines of research and practice. The work should be done in teams in which semantic alignment to IFC is designed as a stable interface and libraries of mappings are published so that they can be reused. Execution outputs need to be presented in an auditable mode, citing each decision to the applicable clause, the supporting evidence and the relevant IFC identifiers. The community should publish open benchmarks, especially between regulatory text, IFC models and gold encodings, implement end-to-end pipelines across multiple codes and project types, benchmark hybrid approaches that combine large language models with classical natural language processing and symbolic execution, and make sure that all final decisions are inspectable and maintained under version control.

Our review is constrained by period covered and based on the sources queried thus the literature and proprietary deployments could be underrepresented. Despite these constraints, the framework and findings provide a workable roadmap and conceptual framework that connects upstream research and modelling with downstream application and facilitate the process of moving promising prototypes into reliable infrastructure in the AEC industry.

Appendix A. Full Keyword Sets for Thematic Classification

Multimodal Information Extraction: Natural Language Processing, NLP, multimodal extraction, OCR, table parsing, diagram understanding, layout analysis, PDF-to-structured, document AI, computer vision in regulations, visual document parsing, multimodal NLP, image-based rules, tabular data extraction, document layout modelling, diagram OCR, vision-language models, image-text alignment, semantic segmentation.

Formalization of Regulatory Text: regulatory formalization, rule extraction, logic encoding, syntax trees, NLP parsing, legal text processing, structured regulations, compliance logic, tokenization, Gherkin, T4R, regulatory modelling, structured rules, legal knowledge extraction, legal informatics, machine-readable law, information extraction, legal document modelling, rule authoring, rule templates, plain-language rules, regulation updates, version control, regulatory change management, delta detection, evolving rules.

Semantic Alignment with BIM/IFC: semantic alignment, IFC mapping, BIM linkage, BIM compliance, rule-to-model alignment, BIM ontology, IFC schema, object-property linking, geometry validation, model interpretation, model checking, digital twin regulation, Level of Detail (LOD), spatial reasoning, BIM data mapping, IDS, Information Delivery Specification, IFC validation rules, information delivery requirements, bSDD, buildingSMART Data Dictionary, MVD, Model View Definition, IDM, Information Delivery Manual, gbXML, CityGML, ISO 19650, BIM interoperability, non-IFC standards, BIM data exchange, information delivery standards.

Integration of Ontologies and Knowledge Graphs: knowledge graph, ontology integration, regulatory ontology, semantic web, RDF triples, OWL classes, SPARQL queries, linked data, graph-based modelling, domain semantics, ontology-driven reasoning, concept hierarchies, data interlinking, bSDD, SKOS, SBVR, business vocabulary and rules, graph enrichment, semantic integration.

Rule Representation and Reasoning: rule formalization, rule-based reasoning, SHACL, SWRL, RDF, OWL, constraint validation, logic programming, inference engine, compliance engine, semantic rules, rule sets, Prolog, logic schema, automated rule evaluation, SBVR, business rules modelling, rule engine, semantic reasoning, ontology reasoner.

Model–Driven Compliance Intelligence: LLM, GPT, BERT, ChatGPT, RAG, Retrieval-Augmented Generation, prompt engineering, zero-shot reasoning, few-shot prompting, fine-tuned transformer, foundation models, large models for regulation, compliance generation, text-to-rule, domain-adapted models, knowledge-augmented generation, semantic retrieval, RAG pipelines.

Explainability and Trust in AI Systems: explainable AI, XAI, model transparency, interpretability, human trust, decision explanation, explainable compliance, rationale generation, rule traceability, visual explanation, regulatory justification, compliance explanation engine, model auditability, human-centered AI, explanation interface, justificatory models, AI bias mitigation, data privacy, regulatory ethics, human-centered assurance, privacy-preserving compliance.

Human-in-the-Loop Approaches: HITL, hybrid AI-human, interactive compliance, manual override, expert validation, user-in-the-loop, participatory compliance checking, semi-automated verification, human feedback loop, collaborative AI, active learning, expert-in-the-loop, human review layer, crowdsourced feedback, human oversight.

Evaluation and Benchmarking: compliance evaluation, benchmark datasets, test suite, precision-recall, F1 score, validation framework, model comparison, compliance metrics, error analysis, test-driven compliance, reproducible testing, validation dataset, performance comparison, evaluation protocol, ground-truth dataset, ACC benchmarking.

Tool Development and Real-World Application: compliance tool, BIM plugin, IFC validation tool, digital permitting, software deployment, commercial integration, regulatory automation system, compliance prototype, rule engine, industry application, field validation, smart permit platform, pilot project, AEC integration, architectural tooling, construction automation, ACC software.

Appendix B. Thematic Papers Matrix

Table 1: Thematic Papers Matrix (Part A)

Authors	Goal	Method	Tools	Domain	Contributions	Theme Mapping	Themes Related (Yes/No)
Zhou, Y., et al. (2022) [30]	Develop a semantic framework that couples BIM data with Design for Safety rules to automate compliance checking	NLP extracts safety clauses, which are formalised as SWRL rules linked to an ontology.	NLPIR, Revit + Dynamo, Protégé with Cellfie, SWRL/OntoGraf	Design-for-Safety regulation in Chinese construction projects	Introduces a purpose-built DFS ontology and demonstrates high recall (95%) and precision (91%) in automated rule evaluation	Rule Representation and Reasoning (0.752)	Yes
Kruiper R., et al. (2024) [31]	Create a national-scale, digital workflow that transforms building regulations from static PDFs into semantically rich, machine-readable knowledge	i-Rec converts 800 regulations into a linked-data graph, delivering semantic search, authoring and analytics tools for compliance	Spar.txt, text-to-KG scripts, RDF, OWL ontologies	UK Building Regulations	Normalises complex, ambiguous regulations into a linked model, enabling superior semantic search and paving path to automated rule execution	Integration of Ontologies and Knowledge Graphs (0.746), Multimodal Information Extraction (0.723), Model–Driven Compliance Intelligence (0.659)	Yes
Y. Li, et al. (2023) [33]	Devise semantic code representation capturing spatial and relational constraints for more accurate automated compliance checks	Five-part model; map codes to IFCOWL; express rules as triples; auto-generate SWRL for reasoning	IFCOWL ontology, OWL framework, and SWRL rule engine	Chinese metro station design regulations	Structured model captures complex spatial constraints, auto-produces ontology-based rules, boosting compliance-checking precision	Rule Representation and Reasoning (0.766)	Yes
L. Jia, et al. (2025) [34]	Combine Semantic Web with IFC data to automate BIM compliance during design.	Five-step ontology process; convert IFC to IFCOWL, map rules, SPARQL queries, validated with Jena.	IFCOWL, OWL ontologies, SPARQL, Jena framework.	Architectural design-phase BIM compliance.	Integrates BIM and codes semantically; reusable ontology and SPARQL templates enhance interoperability.	Rule Representation and Reasoning (0.817)	Yes
J. Anim, et al. (2024) [35]	Enable automated compliance checks with temporal and aggregate reasoning via SHACL-SPARQL on RDF legal norms.	Extend SHACL using SHACL-SPARQL and TVC temporal pattern; encode norms in RDF; validate via SHACL shapes case study.	RDF, SHACL-SPARQL, SPARQL queries, TVC ontology pattern, DLV2 logic reasoner.	Ghanaian petroleum regulation LI 2204 legal compliance.	Adds temporal/aggregate reasoning to SHACL, interoperable RDF approach, provides explanations, adaptable to AEC constraints.	Explainability and Trust in AI Systems (0.703), Human-in-the-Loop Approaches (0.667)	Yes
Y. Wang, et al. (2023) [36]	Automate fire-code compliance via BIM and NLP, emphasising often-neglected spatial geometric relationships.	Pipeline: parse BIM, translate codes to logic via NLP, analyse geometry, output reports; validated on diverse buildings.	Autodesk Revit, Java and Python services in B/S architecture, plus bespoke spatial-analysis algorithm.	Fire-safety compliance for building models.	BIM-NLP framework achieves 97% recall, 91% precision, offering visual modification advice for non-conforming elements.	Semantic Alignment with BIM/IFC (0.713), Evaluation and Benchmarking (0.660)	Yes

Table 2: Thematic Papers Matrix (B)

Authors	**Goal**	**Method**	**Tools**	**Domain**	**Contributions**	**Theme Mapping**	**Themes Related (Yes/No)**
N. Chen, et al. (2024) [37]	Integrate LLM, deep learning and ontology to automate BIM compliance interpretation and checking for greater efficiency.	Lightweight ontology model; DL classifies clauses, LLM extracts structure; outputs translated into SPARQL queries.	Unspecified LLM, deep-learning classifiers, Protégé ontology editor, SPARQL engine, BIM datasets.	Chinese residential building design compliance.	Few-shot LLM and DL minimise annotation, enhance scalability, mapping regulations to ontology reasoning.	Model–Driven Compliance Intelligence (0.697), Multimodal Information Extraction (0.696)	Yes
D. Chen, et al. (2024) [38]	AI-based fire risk assessment and proactive mitigation from raw 2-D blueprints, eliminating reliance on structured BIM models.	Standardises blueprints via CV/OCR, builds connectivity graphs for risk metrics, then generative diffusion redesigns non-compliant layouts.	Python, OpenCV, Tesseract, NetworkX, PyGraphviz, PyTorch-guided diffusion, image-processing libraries for graph extraction and generation.	Fire-safety compliance and risk mitigation for residential building designs derived solely from blueprint image data.	Delivers rapid full-cycle risk assessment (≈ 2s) and mitigation (≈ 40s), robust to noise, extending compliance automation beyond BIM.	Rule Representation and Reasoning (0.741) (Multimodal Information Extraction, Model–Driven Compliance Intelligence)	No
K.W. Johansen, et al. (2024) [39]	Automate guardrail compliance inspections on sites to minimise fall hazards.	Formalise safety rules, analyse UAV/BIM point clouds, Prolog classifies compliance; validated with synthetic and real cases.	UAV-derived point clouds, BIM models, Hough feature extraction, Prolog reasoning engine.	Construction-site safety compliance for temporary guardrail fall-protection systems.	First scalable, explainable guardrail ACC; handles sparse data, 210 k instances; open knowledge base fosters reproducibility.	Rule Representation and Reasoning (0.741)	Yes
Z. Ma, et al. (2024) [40]	Automate BIM model quality compliance via ontology to assure data correctness, consistency, completeness.	Encode standards as OWL/SWRL, convert IFC data to ontology instances, run rule-engine checks.	OWL, SWRL, IFC schema, Jena API, BIMserver platform.	Chinese BIM Quality Standards documents	Introduces BIM-QC ontology and scalable prototype, computerising most requirements and advancing ontology-driven automated quality validation.	Semantic Alignment with BIM/IFC (0.740)	Yes
Q. Ren, et al. (2024) [41]	Automate dam quality compliance by converting natural-language standards into structured, machine-readable rules.	Combine rule syntax parsing with semantic distance to compare regulations and project data.	Rule syntax parser, Word2Vec semantic module, and validation using quality database.	Concrete dam construction quality compliance and regulatory checkin	Hybrid syntax–semantic framework, real-project demonstration, reduced engineer interpretation workload.	Model–Driven Compliance Intelligence (0.659) (Formalisation of Regulatory Text)	No

Table 3: Thematic Papers Matrix (C)

Authors	Goal	Method	Tools	Domain	Contributions	Theme Mapping	Themes Related (Yes/No)
H. Hettiarachchi, et al. (2025) [14]	Publish an open, richly annotated regulatory corpus to catalyse machine-readable rule generation for automated compliance checking.	Extracted 862 self-contained sentences; manually tagged 4,297 entities and 4,329 relations, validating quality throughout.	Used LightTag for annotation, YAML-based BCRL schema, GitHub repository for open distribution.	AEC regulatory texts from England and Finland, prepared for NLP rule-generation research.	Delivers open benchmark enabling entity recognition, relation extraction and LLM training for compliance automation.	Formalisation of Regulatory Text (0.760), Human-in-the-Loop Approaches (0.663)	Yes
E. Nuyts, et al. (2024) [6]	Compare eight ACC techniques interpreting Flemish building rules across five constraint categories to guide tool selection.	Benchmark on uniform IFC datasets, assessing functionality, interoperability, rule expressiveness, and validation reporting.	Solibri, IDS/USBIM.ids, XSD, JSON Schema, OWL + SWRL (Pellet), SPARQL, PySHACL, Revit.	Digital compliance validation for Flemish AECO construction data.	Reproducible evaluation exposes functional gaps, interoperability issues, and recommends hybrid IFC-schema-linked-data frameworks.	Model–Driven Compliance Intelligence (0.659)	Yes
J. Wu, et al. (2025) [42]	Automatically adapt non-compliant BIM designs into code-compliant alternatives while preserving original layout and minimising manual revisions.	Graph-based propagation pinpoints affected elements; Morris sensitivity ranks variables; conditioned Latin hypercube sampling explores feasible design variants.	Revit/Dynamo API, NetworkX graphs, Python, Morris SA, CLHS, weighted Euclidean distance metric.	Architectural spatial compliance (corridor width, room area) under International Building Code during early design stages.	Introduces designer-centred "design healing" workflow linking ACC results to automated, explainable, localised corrections with efficient exploration.	Multimodal Information Extraction (0.668) (Tool Development and Real-World Application, Human-in-the-Loop Approaches)	No
T. Beach, et al. (2024) [43]	Validate open ecosystem architecture for ACC surpassing monolithic BIM-only models.	Implemented modular compliance micro-services via open APIs; tested on school case and industry feedback.	IFC-based BIM, JSON/XML APIs, energy simulation, geometric checks, colour-contrast and BIM-lookup services.	UK non-domestic building compliance for energy, accessibility and fire-safety regulations.	Ecosystem automates 85% rules, scales, auditable, outperforms BIM-only, enabling collaborative distributed ACC.	Human-in-the-Loop Approaches (0.645)	Yes
L. Robaldo, et al. (2023) [44]	Evaluate efficient automated compliance checking directly over RDF data, comparing SHACL rules and DLV2 ASP reasoning.	Formalised normative use case, encoded in both languages, benchmarked on synthetic RDF datasets, local files and SPARQL endpoints.	RDF, SHACL with TopBraid/Jena; DLV2 ASP-core-2 with SPARQL import directives; Apache Jena Fuseki server.	Legal-tech compliance scenarios; RDF-encoded states of affairs for licensing and publication rules.	Finds DLV2 significantly faster than SHACL; validates SPARQL imports; open-sources code; advises scalable parallel workflows.	Explainability and Trust in AI Systems (0.635)	Yes

Table 4: Thematic Papers Matrix (D)

Authors	Goal	Method	Tools	Domain	Contributions	Theme Mapping	Themes Related (Yes/No)
S. Fischer, et al. (2024) [45]	Investigate and propose extensions to IDS for digital building permit and simple code compliance checks.	Analyse current IDS; design schema restrictions; prototype validated in Vienna.	IDS XML schema, IFC models, Solibri Office, BCF workflow.	BIM-based digital permit process and compliance checking in Austria.	Adds related-element, nesting, cardinality rules; maintains simplicity; shares reproducible BIM tests.	Semantic Alignment with BIM/IFC (0.724)	Yes
E. Makisha (2023) [46]	Assess feasibility of translating Russian apartment-building code clauses into machine-readable rules for BIM compliance automation.	Applied RuleML-based algorithm, categorised 356 clauses by translatability, proposed structuring guidelines to resolve ambiguities.	RuleML rule generator, IFC terminology integration, Excel interface for rule authoring.	Russian residential apartment building design-stage BIM compliance.	Found 47% provisions machine-translatable; identified blockers and proposed standards-restructuring strategies for future automation.	Formalisation of Regulatory Text (0.769)	Yes
H. Ying and R. Sacks, et al. (2023) [32]	To create an autonomous and explainable compliance checking system that processes diverse regulatory inputs and integrates with BIM workflows.	Uses LLM agents with Retrieval-Augmented Generation to interpret multimodal regulations	LLM-based agents, Retrieval-Augmented Generation framework, RevitCopilot BIM plugin, multimodal parsing components.	Generic building code compliance checking within architecture, engineering, and construction BIM environments.	Delivers an LLM-driven compliance checker that is autonomous, explainable, interactive, multimodal-capable, and integrated with BIM via RevitCopilot.	Explainability and Trust in AI Systems (0.654)	Yes
T. Bloch, et al. (2023) [47]	Explore graph neural networks for scalable design-review compliance without explicit rule programming.	Convert BIM floorplans to labelled property graphs; train GAT/GCN models; evaluate on synthetic and real datasets.	PyTorch Geometric, GAT, GCN architectures, synthetic graph generator, BIM input.	Residential building accessibility rule checking.	Introduces rule-free GNN workflow, 86–88% accuracy, supplies dataset, captures geometric-relational dependencies.	Tool Development and Real-World Application (0.666)	Yes
R. Zhang, et al. (2022) [48]	Automate transformation of natural-language code sentences into semantic computable requirements for compliance checking.	Semi-supervised RNN extracts semantic relations, builds hierarchical requirement model, evaluated on multi-code corpus.	RNN-based deep learning, semi-supervised training, custom annotations, IFC-linked semantic schema.	IBC, IRC, IEB Code, International Fire Code and ADA/ASHRAE standard	Auto-extract hierarchical requirements, enhancing rule computability and flexibility beyond manual crafting.	Human-in-the-Loop Approaches (0.642)	Yes
D. Al-Turki, et al. (2024) [49]	Automate RASE tagging of regulations to create machine-readable rules for automated compliance checking.	GPT-4o few-shot, fine-tuned, active-learning pipeline converts clauses to YAML, iteratively refined with expert feedback.	GPT-4o, BCRL YAML, active-learning interface.	Building compliance regulations within AEC.	LLM-active-learning RASE annotator; boosts accuracy, slashes manual coding, speeds ACC adoption.	Formalisation of Regulatory Text (0.779), Tool Development and Real-World Application (0.692), Human-in-the-Loop Approaches (0.638)	Yes

Table 5: Thematic Papers Matrix (D)

Authors	Goal	Method	Tools	Domain	Contributions	Theme Mapping	Themes Related (Yes/No)
Z. Zheng, et al. (2022) [50]	Develop knowledge-driven framework to bridge semantic gap and automate rule interpretation for BIM compliance checking.	Build fire ontology, convert IFC to RDF, parse code sentences, Word2Vec aligns terms, repair conflicts, auto-generate SPARQL for compliance checks.	Protégé ontology, Python Word2Vec, ANTLR4 parser, Pellet reasoner, GraphDB SPARQL engine, IFC parser.	Fire-protection requirements in Chinese GB50016; IFC-based plant building BIM models.	Achieved 90% alignment accuracy, five-times faster interpretation, supports implicit attributes and complex rules via SPARQL.	Integration of Ontologies and Knowledge Graphs (0.779)	Yes
S. Madireddy (2025) [51]	Integrate LLMs with BIM for semi-automated code compliance, reducing manual rule programming in design verification.	LLM converts regulations to Python via prompt-engineering; Revit executes; error-feedback refines scripts automatically.	ChatGPT, Claude, Gemini, LLaMA; Revit API; Python; iterative prompt templates.	BIM compliance for residential and office models, verifying dimensions, materials, and object placement.	Cuts checking time, auto-generates executable rules, improves geometric and attribute detection, adaptable across varied regulations.	Formalisation of Regulatory Text (0.738), Multimodal Information Extraction (0.694), Evaluation and Benchmarking (0.694), Model–Driven Compliance Intelligence (0.684)	Yes
L. Jiang, et al. (2022) [1]	Integrate multiple ontologies to resolve semantic heterogeneity for automated BIM code compliance checking.	Develop four ontologies, map terms, merge, apply SWRL reasoning to generate compliance reports; five-step ontology roadmap.	OWL ontologies, Protégé, IFC/ifcOWL, SWRL rule engine.	Chinese building design codes applied to BIM structural and design specifications.	Introduces multi-ontology fusion; transparent grey-box reasoning; improves compliance accuracy and efficiency over manual checks.	Rule Representation and Reasoning (0.738)	Yes
H. Liu, et al. (2023) [52]	Accelerate validation of MVDXML rules for BIM data exchange.	MVDLite merges templates and rules into chains, employs round-trip search, filtering, deep caching to avoid redundant graph visits.	MVDXML, IFC schema, steparser implementation; benchmarks vs XBIM and Python-MVDXML.	IFC data-exchange compliance under buildingSMART Model View Definitions.	Achieves 3–14× speed-up and 40% runtime cut on large rule sets, foundational for faster BIM interoperability checks.	Semantic Alignment with BIM/IFC (0.758)	Yes
O. Doukari, et al. (2022) [53]	Bottom-up, object-centred ACC framework to boost quality, efficiency and scalability when certifying BIM object libraries.	Four-stage pipeline: parse rules, prep object data, execute if-then XML rules with simulated-annealing optimisation, report results.	XML rule sets, IFC objects, Revit plug-in, simulated-annealing algorithm, expert workshops for rule formalisation.	Food-service equipment and fire-door BIM objects validated against FCSI and building-code standards.	Cuts manual checking by $\approx$ 125 minutes/object; hierarchical rule decomposition; extensible architecture for future IFC and open-standard integration.	Evaluation and Benchmarking (0.704), Explainability and Trust in AI Systems (0.686)	Yes

Table 6: Thematic Papers Matrix (F)

Authors	**Goal**	**Method**	**Tools**	**Domain**	**Contributions**	**Theme Mapping**	**Themes Related (Yes/No)**
J. Fauth, et al. (2025) [54]	Automate building-permit reviews via ontology-driven, containerised data exchange and compliance checks.	LOT layered ontologies, SHACL rules, ICDD containers, RDF querying; demonstrated in German case study.	OWL/Protégé, SHACL, RDF/SPARQL store, ISO 21597 ICDD, web review prototype.	German municipal permitting; BIM-based legal and technical compliance verification.	Unified ontology-process framework, automated notifications, semantic interoperability, adaptable across jurisdictions.	Integration of Ontologies and Knowledge Graphs (0.785)	Yes
F. Yang (2024) [55]	Automate translating building-code text into machine-readable logic using GPT prompt engineering.	Two-module system; prompts convert text to logic clauses; evaluated on code and crash datasets.	GPT LLM, tailored prompts, Python implementation, logic-programming output format.	IBC 2015 Fire Protection clauses plus crash-report corpus for cross-domain testing.	97% precision with 2% training data; 28% faster than rule-based; released benchmark dataset.	Tool Development and Real-World Application (0.691), Evaluation and Benchmarking (0.664)	Yes
S. Zhu, et al. (2025) [56]	Automate BIM quality checks via knowledge graphs, minimising manual effort and accommodating evolving standards.	Four-stage flow: convert standards to KG, extract Revit data, run rules, generate visual reports.	SQL dynamic rule base, KG tech, Revit API, C# plug-in, in-Revit visualisation.	Xuzhou Metro Line 6 BIM (2496 components, multi-disciplinary) under Chinese standards.	Dynamic KG removed errors (15.95%), customisable rules, real-time localisation.	Semantic Alignment with BIM/IFC (0.713)	Yes
A. Karmakar, et al. (2025) [57]	Address IFC topology gaps via semantic enrichment enabling complex ACC rule evaluations.	Workflow: classify new elements, create associations, compute geometry, enrich, execute Python rules.	IFC4 schema, Revit modelling, Python checker, geometric reconstruction algorithms.	Mumbai residential BIM; UDCPR sunshade width check on 70 slab elements.	Multi-layer enrichment removed manual preprocessing, enabled derived-attribute rule checks.	Tool Development and Real-World Application (0.689), Evaluation and Benchmarking (0.654)	Yes
Z. Kabzhan, et al. (2025) [58]	Ontology-NLP approach to analyse construction regulations, improving consistency and paving the way for digital compliance.	Ontology modelling + SpaCy-based EKG-LP parsing to build subject-predicate-object profiles and flag contradictions.	BFO, IFC, SKOS ontologies; EKG-LP NLP engine; RDF triple representation.	14 Kazakhstani building codes (≈242 k words) in the AEC sector.	F1 0.86 duplicate/contradiction detection; machine-readable semantic profiles adaptable for BIM-linked ACC.	Formalisation of Regulatory Text (0.750), Integration of Ontologies and Knowledge Graphs (0.731), Multimodal Information Extraction (0.707)	Yes
V. Schuk, et al. (2022) [59]	Define specifications for infrastructure ACCC tools and expose research gaps.	Systematic review; developed classification; evaluated BPMN/DMN, RASE, graph modelling.	BPMN, DMN, RASE, graph databases, IFC-RAIL, knowledge-based engineering systems.	Railway guidelines (DB AG Ril 800); 943 rules; extension possible to road transport.	Faceted framework; 85% parametric rules BPMN/DMN-representable; paves way for standardised infrastructure ACCC.	Integration of Ontologies and Knowledge Graphs (0.745), Tool Development and Real-World Application (0.694)	Yes

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